

MANOBHI SRI RAM

Generative AI Engineer · Agentic AI Systems · LLM Orchestration · RAG Pipelines
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PROFESSIONAL SUMMARY

Generative AI Engineer with 1 year of production impact — sole AI technical lead at Ascendion, architecting and delivering two enterprise Agentic AI systems now adopted company-wide across 1,500+ employees in 12 countries. Led cross-functional delivery with DevOps, IT security, and senior management; ran daily standups and sprint reviews; presented milestone decisions directly to company directors. Built a multi-model facial recognition pipeline (97.3% accuracy, 50K+ images, retrieval from 8 minutes to 0.4 seconds, eliminating an entire manual media processing workforce) and a LangGraph multi-agent mobility platform that replaced a 13-hour manual coordination workflow with zero human intervention — becoming Ascendion's standard operating platform for executive mobility globally. Core expertise: Agentic AI system design, LLM orchestration, Retrieval-Augmented Generation (RAG), prompt engineering, model fine-tuning (LoRA/QLoRA), and LLM evaluation — deployed in production on AWS and Azure. Targeting senior AI engineering roles demanding full technical ownership, AI system design, and measurable production impact.

WORK EXPERIENCE

ASCENDION · Chennai, TN

Aug 2025 – Present

AI Engineer — Generative AI & ML | Technical Lead

Project 1: Enterprise AI Media Intelligence Platform — Facial Recognition & Semantic Vector Search

Company-wide facial recognition and semantic media retrieval platform — indexes enterprise event photos across 1,500+ employees globally and enables instant identity-based vector search, replacing an entire manual image processing and segregation workforce.

- Sole AI technical lead — **coordinated cross-functional delivery across DevOps, IT security, and senior management**; ran daily standups and sprint reviews; owned full AI system design from architecture to production deployment; presented milestones directly to company directors.
- Architected multi-model facial recognition pipeline (DeepFace/FaceNet512, RetinaFace, MTCNN) with adaptive preprocessing for group photos, low-light, and varied angles — **97.3% match accuracy across 50K+ indexed enterprise images** in production; applied prompt engineering for dynamic query handling at inference.
- Designed Pinecone vector search DB storing 512-dimensional facial embeddings with cosine similarity — reduced retrieval from **8+ minutes to 0.4 seconds**; applied latency optimisation via concurrent ThreadPoolExecutor batch inference with zero downtime on AWS EC2.
- Delivered **94% reduction in post-event media processing (13 hours → 45 minutes)** — eliminated the entire manual image segregation and processing workforce; platform adopted daily by 350+ employees, saving an estimated 200+ person-hours of manual labour per month across the organisation.
- Engineered privacy-first biometric architecture with encrypted inference pipelines, zero persistent selfie storage, session tokens, and role-based controls — **cleared full enterprise IT security and compliance review**; implemented production ML observability via CloudWatch with end-to-end pipeline monitoring.

Project 2: Leadership Mobility Command Center — Generative AI + Agentic AI Enterprise Operations Platform

Agentic AI platform for fully autonomous executive travel coordination across Ascendion's global operations — tracks live flight telemetry, optimises ground routes, broadcasts real-time updates, and manages driver dispatch end-to-end with zero manual intervention across all offices in 12 countries.

- Sole AI technical lead — **led cross-functional coordination across DevOps, IT, driver operations, and executive admin teams**; ran daily standups and sprint reviews; presented live demos to senior directors at every delivery phase.
- Designed and deployed LangGraph multi-agent orchestration system (flight intelligence + ground transport agents, shared trip-state memory, autonomous lifecycle management) — replaced all fragmented manual workflows with a **unified Agentic AI command centre adopted company-wide across 1,500+ employees in 12 countries**; now Ascendion's global standard operating platform for executive mobility.
- Built production Retrieval-Augmented Generation (RAG) pipeline (LangChain · Claude 3.5 Sonnet · Qdrant) on 5,000+ vectorised trip records — eliminated all manual trip coordination, follow-ups, and status tracking; **estimated 300+ coordinator hours saved per month** across all global offices.
- Integrated Aviationstack real-time flight telemetry, Google Maps API, Microsoft Teams broadcasting, and Twilio/Firebase GPS sync — **fully autonomous end-to-end operation with zero manual dispatch since launch**; implemented LLMops in production with full observability via Langfuse for every agent decision.

PROJECTS

AI Meeting Intelligence Platform · LangGraph Multi-Agent · RAG · Production

Personal Project | Sole Architect & Developer | 2024

Production-deployed Agentic AI system that transcribes meeting audio, extracts structured tasks via LLM chains with prompt engineering, retrieves context via Retrieval-Augmented Generation (RAG), and autonomously routes action items across Slack, Teams, and email — eliminating all manual post-meeting follow-up.

- Designed LangGraph multi-agent system with conditional state-based routing — agent graph dynamically selects notification channels (Slack, Teams, email) based on task priority, employee context, and meeting state; applied AI system design principles to eliminate all hardcoded workflow logic.
- Built LangChain + Groq (LLaMA 3) extraction chains with structured output parsers and prompt engineering — **~90% task extraction accuracy across 500+ tasks** with sub-2s inference latency; latency optimisation via async chain execution; extracted fields include owner, deadline, priority, and dependencies.
- Implemented Pinecone vector search for three RAG use cases: semantic transcript search, conversational Q&A over meeting history, and cross-meeting duplicate task detection — with LLM evaluation metrics tracked per pipeline run.
- Deployed on AWS with FastAPI backend — real-time SaaS dashboard with overdue tracking, completion analytics, and full activity timeline; CRM-ready integration layer enabling **enterprise drop-in deployment with zero pipeline restructuring**.

MedAssist — Production Clinical Triage Agent

Personal Project | Sole Architect & Developer | 2024

Production-grade local-inference Agentic AI triage system for regulated Indian healthcare environments where patient data cannot leave institutional servers — accepts symptoms in plain English and returns a fully explainable triage decision with severity classification, red flag warnings, drug interaction checks, and citations traceable to WHO and ICMR guidelines.

- Fine-tuned Phi-3 Mini 3.8B on 15,000 medical QA examples using LoRA — **training loss reduced from 1.92 to 1.21**; genuine domain adaptation to clinical triage language; latency optimisation via Ollama local inference with zero cloud dependency.
- Built 5-node LangGraph agent with conditional routing — HIGH/EMERGENCY cases trigger parallel OpenFDA drug interaction checks; LOW/MEDIUM skip it (**~30% latency reduction**); full AI system design with Pydantic schema validation on every node output; missing severity field triggers automatic corrective retry.
- Hybrid vector search + full-text search in LanceDB (fully local, zero cloud dependency) with semantic chunking and custom Reciprocal Rank Fusion — **context precision improved from 0.65 to 0.85** vs. fixed chunking; Production ML observability via Langfuse with per-decision audit trail for regulatory review.
- LLM evaluation on curated clinical QA pairs via Ragas: **faithfulness 0.92, answer relevancy 0.90, context precision 0.85, context recall 0.87**; DeepEval CI/CD via GitHub Actions blocks any push degrading faithfulness below 0.60 or causing severity misclassification — treated as a patient safety gate, not a performance metric.

TECHNICAL SKILLS

Generative AI / LLM: LangChain · LangGraph · CrewAI · AutoGen · Agentic AI · RAG · Retrieval-Augmented Generation · Prompt Engineering · Fine-Tuning (LoRA/QLoRA) · Hugging Face · OpenAI API · Anthropic API (Claude) · LLM Orchestration · AI System Design

ML / LLM Evaluation: LLM Evaluation · Deep Learning · NLP · Ragas · DeepEval · LangSmith · Langfuse · Weights & Biases · MLflow · Production ML · MLOps in Production

Vector Search / DBs: Vector Search · Pinecone · Qdrant · LanceDB · FAISS · Chroma · Weaviate

Observability: Langfuse · LangSmith · CloudWatch · Weights & Biases · MLflow · CI/CD · GitHub Actions

Languages: Python · SQL · JavaScript

Backend / APIs: FastAPI · Flask · Node.js · REST APIs · WebSockets

Cloud / Infra: AWS (EC2 · S3 · Bedrock · CloudWatch · IAM) · Azure · Docker · Firebase · GitHub Actions

Web (Supporting): React · Next.js · PostgreSQL · MongoDB

CERTIFICATIONS

- **Microsoft Certified: Azure AI Developer Associate** — Microsoft · 2024
- **AWS Certified Cloud Practitioner** — Amazon Web Services · 2024

EDUCATION

Vellore Institute of Technology (VIT), Chennai

B.Tech, Computer Science & Engineering · CGPA: 8.39 / 10

Aug 2021 – Aug 2025